



Don't Be A Loser!

You don't know
how much you
don't know.

Read

electronics
FOR YOU

to keep up-to-date
with the tech world



To subscribe,
visit efy.in or
contact us at: 011-4059660
Email: support@efy.in

OR

Scan
This
Code



Also available at all leading magazine outlets

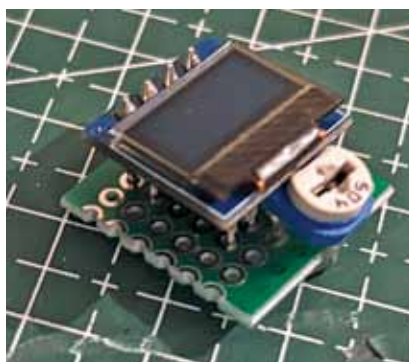


Fig. 5: Game Boy with an SMD screw potentiometer for a more compact design

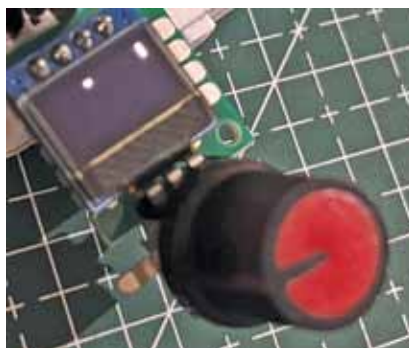


Fig. 6: Game Boy running the Pong game

Construction and testing

After designing the circuit and writing the source code, the next step is to upload the code to the ATtiny85. Once the upload is complete, proceed with the hardware assembly. For this, use a zero PCB and carefully cut it to achieve the desired compact size. Mount the ATtiny85, OLED display, and other components, then solder them together while fitting everything within a compact space of approximately 2.5cm to 2.7cm. This ensures an ultra-compact and well-integrated design.

To further reduce the size, all components are arranged efficiently on the PCB to achieve a highly dense layout. For the control input, a standard potentiometer can be replaced with an SMD screw-type potentiometer, making the overall design even smaller and more practical.

Fig. 4 shows the soldering process of the Game Boy components, while Fig. 5 shows the

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to
<https://sourcecode.electronicsforu.com>
to access and download source code
and other resources for this project

assembled Game Boy using an SMD screw potentiometer for a more compact design.

For testing, connect the solar cell to the solder pads (PAD1 and PAD2) on the assembled Game Boy PCB. This connection enables completely battery-free operation, allowing the device to directly harvest solar energy.

Under sufficient lighting, the system runs continuously from the solar cell. In low-light conditions, the supercapacitor can be charged within a few seconds, allowing gameplay to resume without interruption. Fig. 6 shows the Game Boy running the Pong game.

Power consumption and runtime

In this design, the ATtiny85 typically consumes 3mA to 5mA at 8MHz. The 0.49-inch (1.2cm) 64 × 32 SSD1315 OLED display draws 5mA to 10mA, depending on the number of active pixels. Therefore, the total system current during gameplay is around 10mA to 20mA.

Using a 2F, 2.7V supercapacitor, the usable stored energy, from approximately 2.7V down to 2.0V, is about 3.3 joules. With an average power consumption of around 30mW to 40mW, this stored energy is sufficient to run the system for a short duration. In summary, the Game Boy can operate completely without a battery, powered either directly by the solar cell or by the energy stored in the supercapacitor. **EFY**

Ashwini Kumar Sinha, an IoT and AI enthusiast, is Tech Journalist at EFY.